Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/08/2005
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Project Title:	Gravity response mechanisms of lateral organs and the control of plant architecture		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mullen, Jack (Indiana University)		
Grant/Contract No.:	NNA04CC55G		
Performance Goal No.:			
Performance Goal Text:			

Using Arabidopsis as a model, we have found that this change in orientation is not simply due to the onset of gravitropic competence, as non-vertical lateral roots are capable of both positive and negative gravitropism. Thus, the horizontal growth of the new lateral roots is determined by what is called the gravitropic set-point angle (GSA). In Arabidopsis shoots, rosette leaves and inflorescence branches also display GSA-dependent developmental changes in their orientation. This developmental control of GSA in lateral organs of Arabidopsis provides a useful system for investigating components involved in regulating the directionality of tropistic responses. The goal of our proposal is to use a combination of genetic, molecular, and physiological approaches to better understand the role of gravity in guiding whole plant development. We propose to identify Arabidopsis mutants that have altered lateral root orientation while maintaining normal primary root orientation. Likewise, we will identify mutants that have altered leaf and/or branch

Task Description:	orientation. We have already identified several such gsa mutants representing four loci. We propose to continue screening for new gsa mutants, clone and characterize the mutant and wild-type alleles of the genes already identified, and determinee the function of the gene products. The molecular and physiological analyses of the gsa mutants will provide new insight into the basis of GSA regulation and, thus, a better understanding of the physiological and molecular mechanisms by which plants use gravity to control their architecture. To discover and characterize genes that are required for controlling the orientaion of lateral organs during plant development. The molecular and physiological analyses of the gsa mutants will provide new insight into the basis of GSA regulation and, thus, a better understanding of the physiological and molecular mechanisms by which plants use gravity to control their architecture. Since the bulk of a plant consists of lateral organs, our studies will contribute to our understanding of the fundamental developmental biology of plant form and may eventually lead to the development of improved plant varieties for growth on future long-term space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Since the bulk of a plant consists of lateral organs, our studies will contribute to our understanding of the fundamental developmental biology of plant form and may eventually lead to the development of improved plant varieties for high-density agricultural.
Task Progress:	We have identified three mutations that specifically affect the gravity-dependent orientation of lateral roots that represent different genes. In addition, we have found 2 other likely mutants that are currently undergoing further analysis to determine if they represent novel loci. We have also confirmed the identification of three genes that specifically affect the gravity-dependent orientation of lateral organs in Arabidopsis shoots. Mutations in these latter 3 genes lead to plants with more vertical leaves and lateral branches. Several additional mutant plants with vertical leaf angles have been isolated and are currently being tested to determine if they represent different genes. We have genetically mapped one of the shoot mutants, mgsa4, to a small region of chromosome 4 that spans 2 BAC clones. These BACs have been subcloned as 10 to 15 KB cosmids and are being transformed into the mgsa4 mutant in an attempt to rescue the mutant phenotype. Once transformed plants are identified that have wild-type leaf orientation, we should be able to narrow the mutant gene to several candidates. These candidate genes will be sequenced to determine which gene represents MGSA4. We have also been making progress mapping the root orientation mutant, mgsa3. However, the map interval is still too large to proceed with rescue attempts. We have also been making good progress with physiological characterization of the GSA responses of the mutant and wild-type plants. Papers describing these studies are being prepared for submission. An important discovery made during these studies has come through studies of how the gravity response of lateral roots is under apical control of the primary root. Our experiments have shown that a transmissible signal coming from the primary root apex alters the growth of lateral roots upon gravistimulation of the plant. Our results indicate that while auxin is involved, another unidentified signal is involved. We have also found that the auxin gradient that develops upon reorientation of primary roots, also develops transiently during reorientation of lateral roots. However, when a lateral root is growing at its set point angle, no auxin gradient can be detected. Thus, the auxin gradient develops only when an organs orientation is changed from its set point and a homeostatic uniform distribution is maintained at the set point.
Bibliography Type:	Description: (Last Updated: 08/11/2025)
Articles in Peer-reviewed Journals	Mullen, J.L., Wolverton, C., Hangarter, R.P. "Apical Control, Gravitropic Signaling, and the Growth of Lateral Roots in Arabidopsis" Adv Space Res , Jan-2005